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Chapter 2: Lines and Planes in $\mathbb{R}^3$ — Summary

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Abstract

This summary presents the core concepts of lines and planes in $\mathbb{R}^3$. Lines are described using a point and a direction vector, while planes are defined by a point and a normal vector, each with multiple equivalent representations. It covers intersections, distances, and angles between lines, planes, and points, providing key formulas for practical computations. The summary consolidates essential vector-based techniques for analyzing spatial relationships in three-dimensional space. The material follows [1] and [2].

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1 Lines in $\mathbb{R}^3$

A line in space is determined by a **point** $P_0(x_0, y_0, z_0)$ and a **direction vector** $v = (v_1, v_2, v_3)$, as illustrated in Figure 1.

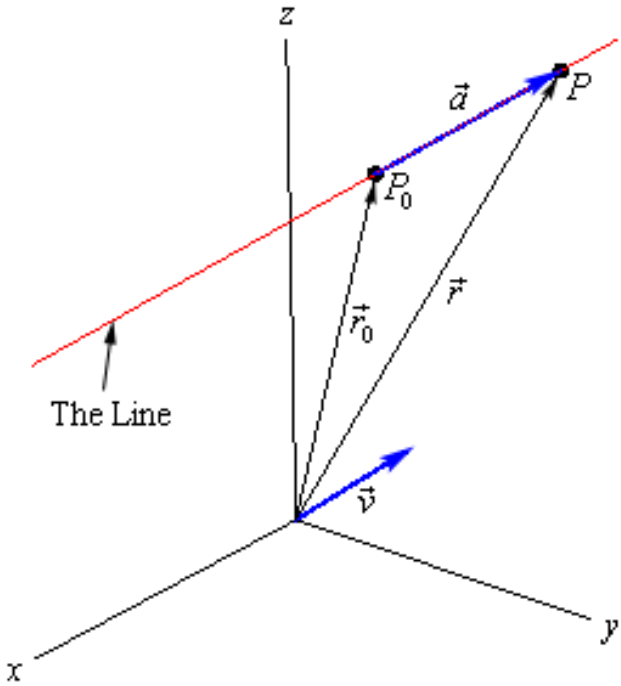

Figure 1: A line in $\mathbb{R}^3$ determined by a point P_0 and a direction vector v.

1.1 Three equivalent forms

1.1.1 Vector form

$$r_1 = r_0 + \lambda v, \quad \lambda \in \mathbb{R}$$

1.1.2 Parametric form

$$\begin{cases} x = x_0 + \lambda v_1 \\ y = y_0 + \lambda v_2 \\ z = z_0 + \lambda v_3 \end{cases}$$

1.1.3 Canonical form

$$\frac{x - x_0}{v_1} = \frac{y - y_0}{v_2} = \frac{z - z_0}{v_3} = \lambda$$

Requires all $v_i \neq 0$.[†]

The representation is not unique — *any* point on the line can serve as the base point, and any nonzero scalar multiple of v gives the same line.

[†]If a component $v_i = 0$, the corresponding fraction is omitted and replaced by the condition $x_i = x_{i,0}$ instead. For example, if $v_3 = 0$, the canonical form becomes $\frac{x-x_0}{v_1} = \frac{y-y_0}{v_2}$ together with $z = z_0$.

2 Planes in $\mathbb{R}^3$

A plane is determined by a **point** $R_0(x_0, y_0, z_0)$ and a **normal vector** $n = (a, b, c)$ perpendicular to the plane. $\langle u, v\rangle = \sum_{i=1}^{3} u_i v_i$ denotes the dot product of vectors u and v.

2.1 Three equivalent forms

2.1.1 Vector form

$$\langle n, r - r_0\rangle = 0$$

2.1.2 General form

$$ax + by + cz = d, \quad \text{where } d = ax_0 + by_0 + cz_0$$

The general form is obtained from the vector form by expanding the dot product:

$$\begin{aligned}
\langle n,\, r - r_0\rangle &= 0 \\
a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0 \\
ax - ax_0 + by - by_0 + cz - cz_0 &= 0 \\
ax + by + cz &= ax_0 + by_0 + cz_0 \\
&= d.
\end{aligned}$$

2.1.3 Vector parametric description

$$r_0 + \lambda u + \mu v, \quad \lambda, \mu \in \mathbb{R}$$

where u, v are two linearly independent vectors lying in the plane.

To convert parametric $\to$ general: find $n = u \times v$ (the cross product of the two direction vectors), then substitute the support point to get d.

Three noncollinear points A, B, C also determine a plane via:

$$\begin{vmatrix} x - x_0 & y - y_0 & z - z_0 \\ x_1 - x_0 & y_1 - y_0 & z_1 - z_0 \\ x_2 - x_0 & y_2 - y_0 & z_2 - z_0 \end{vmatrix} = 0$$

3 Intersections

3.1 Two lines

Two lines in $\mathbb{R}^3$ fall into one of the following configurations:

i. **Intersecting** — exactly one point in common;

ii. **Parallel** — direction vectors are proportional, no intersection;

iii. **Identical** — infinitely many points in common (same line);

iv. **Skew** — not parallel and do not intersect (only possible in $\mathbb{R}^3$).

Lines $\ell_1 = r_1 + \lambda_1 v_1$ and $\ell_2 = r_2 + \lambda_2 v_2$ intersect iff there exist λ_1, λ_2 such that:

$$r_1 + \lambda_1 v_1 = r_2 + \lambda_2 v_2$$

This gives a system of 3 equations; check consistency in all three coordinates.

3.2 Line and plane

Substitute the parametric line $\ell = r_0 + \lambda v$ into the plane equation $ax + by + cz = d$ and solve for λ. Then substitute back to get the intersection point.

→ If $\langle n, v \rangle \neq 0$: unique solution exists — the line meets the plane at exactly one point.

→ If $\langle n, v \rangle = 0$ and the point r_0 does not satisfy the plane equation: no solution — the line is parallel to the plane.

→ If $\langle n, v \rangle = 0$ and r_0 lies on the plane: all $\lambda \in \mathbb{R}$ satisfy it — the line lies entirely in the plane.

3.3 Two planes

Two non-parallel planes intersect in a **line**. Solve the system of two plane equations, express two unknowns in terms of the third (assign it parameter λ), and write the result in vector form.

4 Distances

Table 1 summarises the main distance formulas covered in this section.

4.1 Point to line

Let R_1 be the point, $\ell = r_0 + \lambda v$ the line:

$$\Delta = \frac{|(R_1 - r_0) \times v|}{|v|}$$

4.2 Point to plane

Let R_1 be the point, Π a plane with normal n, and R_0 any point on Π:

$$\Delta = \frac{|\langle R_1 - R_0, n\rangle|}{|n|} \tag{1}$$

Example 4.1. Find the distance from the point $R_1 = (1, 2, 3)$ to the plane $2x + y - z = 4$. Taking $n = (2, 1, -1)$ and $R_0 = (2, 0, 0)$ on the plane, Equation (1) gives:

$$\begin{aligned}
\Delta &= \frac{|\langle R_1 - R_0,\, n\rangle|}{|n|} \\
&= \frac{|\langle(-1,\, 2,\, 3),\, (2,\, 1,\, -1)\rangle|}{\sqrt{4+1+1}} \\
&= \frac{|-2+2-3|}{\sqrt{6}} \\
&= \frac{3}{\sqrt{6}} \\
&= \frac{\sqrt{6}}{2}.
\end{aligned}$$

4.3 Two parallel lines

Check parallelism: $v_1 \times v_2 = 0$ (equivalently, v_1 and v_2 are linearly dependent). Distance = distance from any point on ℓ_1 to ℓ_2 (use the point-to-line formula).

4.4 Two skew lines

Lines that are not parallel and do not intersect. Formula:

$$\Delta = \frac{|\langle v_1 \times v_2, r_2 - r_1\rangle|}{|v_1 \times v_2|}$$

4.5 Line and parallel plane

First verify parallelism: $\langle n, v \rangle = 0$. Distance = distance from any point on the line to the plane (use Equation (1)).

4.6 Two parallel planes

First verify: $n_1 = \alpha n_2$ for some scalar α. Distance = distance from any point on one plane to the other plane.

Configuration	**Formula**
Point to line	$\dfrac{\lvert (R_1 - r_0) \times v \rvert}{\lvert v \rvert}$
Point to plane	$\dfrac{\lvert \langle R_1 - R_0,\, n \rangle \rvert}{\lvert n \rvert}$
Two parallel lines	reduce to point-to-line
Two skew lines	$\dfrac{\lvert \langle v_1 \times v_2,\, r_2 - r_1 \rangle \rvert}{\lvert v_1 \times v_2 \rvert}$
Line and parallel plane	reduce to point-to-plane
Two parallel planes	reduce to point-to-plane

Table 1: Summary of distance formulas for configurations in $\mathbb{R}^3$.

5 Angles

5.1 Between two lines

The angle ϕ is always taken as **acute**:

$$\cos\phi = \frac{|\langle v_1,\, v_2\rangle|}{|v_1|\cdot|v_2|}$$

Theorem 5.1 (Perpendicularity of lines)**.** *Two lines with direction vectors v_1 and v_2 are perpendicular if and only if $\langle v_1,\, v_2\rangle = 0$.*

Proof. The angle formula gives $\cos\phi = \dfrac{|\langle v_1,\, v_2\rangle|}{|v_1|\cdot|v_2|}$. Perpendicularity means $\phi = 90°$, so $\cos\phi = 0$, which holds if and only if $\langle v_1,\, v_2\rangle = 0$. ♠

5.2 Between a line and a plane

The angle ϕ is between the line and its projection onto the plane:

$$\sin\phi = \frac{|\langle v, n\rangle|}{|v|\cdot|n|}$$

where n is the normal to the plane. Equivalently: $\phi = 90° - \beta$, where β is the angle between v and n.

5.3 Between two planes

The angle equals the angle between the normal vectors:

$$\cos\phi = \frac{|\langle n_1,\, n_2\rangle|}{|n_1|\cdot|n_2|}$$

Proposition 5.2 (Orthogonality of planes)**.** *Two planes with normal vectors n_1 and n_2 are* ***orthogonal*** *if and only if $\langle n_1,\, n_2\rangle = 0$.*

Proof. From the angle formula, $\cos\phi = \dfrac{|\langle n_1,\, n_2\rangle|}{|n_1|\cdot|n_2|}$. The planes are orthogonal when $\phi = 90°$, i.e. $\cos\phi = 0$, which occurs if and only if $\langle n_1,\, n_2\rangle = 0$. ♠

References

[1] P. M. Protter and C. B. Morrey, *College Calculus with Analytic Geometry*, 2nd ed., Addison-Wesley, Reading, MA, 1977.

[2] *Algebra I — Lecture Notes*, University of Primorska FAMNIT. Available at: https://e.famnit.upr.si/pluginfile.php/747259/mod_resource/content/19/Algebra%201%20lecture%20notes_V3.pdf